

[24]: *#Based on the Random Forest model's feature importance, the most significant variables affecting car prices are primarily enginesize and curbweight, whi*

```
[25]: #Hyperparameter Tuning
param_grid = {
    "n_estimators": [100, 200],
    "max_depth": [5, 10, None],
    "min_samples_split": [2, 5]
}
grid_search = GridSearchCV(
    RandomForestRegressor(random_state=42),
    param_grid,
    cv=5,
    scoring="r2",
    n_jobs=-1
)
grid_search.fit(X_train, y_train)
```

[25]:

GridSearchCV ⓘ ?

best_estimator_: RandomForestRegressor

RandomForestRegressor(max_depth=10, n_estimators=200, random_state=42)

RandomForestRegressor ⓘ

```
[26]: print("Best Parameters:", grid_search.best_params_)
print("Best Score:", grid_search.best_score_)

Best Parameters: {'max_depth': 10, 'min_samples_split': 2, 'n_estimators': 200}
Best Score: 0.8911213781660681
```